

Evergreen Project: Technical Paper

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The Lopez Urban farm, located on West Mission Boulevard in Pomona, is a project not only to provide vegetables and greenery to their local community, but also to maintain sustainability as an agricultural producer. These benevolent ambitions are reliant on volunteer work, and we wish to contribute to this community project through our skills as Computer scientists. There were many issues we could potentially address, modernizing their irrigation/watering systems, creating a way to log and calculate the weight of crops given away/paid for, and even building an online tour of the farm via scannable QR codes and web pages. Of course, many of these projects would have limitations, considering the farm is off the city's power grid, and there is no personal computer or even a laptop central to the farm. Ultimately, we landed upon two questions to answer: How do we assist in seed management? And, how do we provide this service to the limited technology on site?

We decided that it would be best to represent crops as their own class. Utilizing private fields such as the number of seeds left, the day planted, the day harvested, average crop time, growing season, etc, with public setters and getters. Which would be arranged with the potential for expansion, in case we wanted to add other statistics later on, such as ideal humidity, or even an internally calculated crop period. Though for the most part, we would arrange the information of each crop to be inputted manually, otherwise it would require agricultural research/expertise. These crops could be stored in a simple array (preferred over linked data structures due to RAM), and their information pulled out and displayed onto a table to the user, with buttons to allow users to create, edit, and delete crops through a GUI. Most imperative and perious to the project was the decision to program in Spring-Boot. It is a Java-based framework that would allow us to work within a familiar programming language while being able to run our application on IOS devices. The latter of those two boons is the most important, considering the only

technology available to the Lopez Urban Farm is a single iPad. The goal was to get something minimally functional and fully presentable to the director of Operations, Bianca Ustrell, as soon as possible, but still lay a quality foundation for potential expansions by other programmers.

There were no roles or assigned tasks, no structured means of management or delegation. Instead, all we had was the project proposal and an image of a table provided to us by Bianca Ustrell. Most of us appeared to prioritize other work, along with simply waiting for someone else to give us a task or remind them that the project even exists. Upon actually working to complete the project, we used countless YouTube videos alongside hours spent on Stack Overflow trying to comprehend what I needed to do in order to get a semi-working product done by the deadline. We were able to finish the Saving/Loading feature, along with the GUI, through this methodology. There was heavy use of base, Java/programming knowledge from our college education to establish the base program, and the crop class object. Notably, there was no special or central use of ADTs, given that they were not relevant to the functions of our program. For a long while, there was no means of demoing our project, and as such, the only means of testing for the effectiveness of the crop class was through Junit testing. Which was not expanded into edge cases, but used to test the class and its fields work within minimum expectations.

At the time of writing this, the source code accessible via our GitHub repository has a more or less functioning Minimum Viable Product (MVP). The application runs, presents a GUI to the user, you can add crops, edit crops, delete crops, load and save from a file, and the destination of said JSON file is obscured to the viewer. There may not be any data hashing to secure the data in the file. This is much closer to the MVP than imagined, considering the little amount of time spent working on it. Certain ideal features are missing, such as being able to manually load multiple files, being able to easily locate/move files, and being able to explore the

user's directories for crop.json files. These features will be important if we develop any features that depend on previous data, perhaps even keeping track of average crop periods over multiple years. There has been no testing as to how/if this application currently works on an iPad, fortunately, this can be done easily due to a team member having similar hardware.

Though we have currently reached MVP status, we did not quite meet the deadline for the program/presentation itself. Evidently, given our expedited development, this team was fully capable of delivering an impressive project within the allotted time. A terrible mismanagement of time and work ethic is the ultimate root cause of the rushed nature of the project. Thus, we can consider the project "submitted late" to better reflect the reality of the situation. The team will do its best to restructure its hierarchy/interpersonal sense of responsibility. Most importantly, we will implement some form of structure to ensure none of us feel lost as to the nature of the project and what we may do to contribute. Regular meetings via voice call, to-do lists, and perhaps specified roles will be implemented to ensure progress continues at a steady, viable rate. We do indeed have plans for the future, and will continue to work on this project despite it no longer providing class credit. There are no current deadlines set for the stage of completion/expansion, nor does Bianca UStrell seem especially concerned over when our program comes to fruition. Most important would be ensuring the application works on Bianca Ustrell's device, along with fine-tuning the program to fit her needs. It would be rather convenient if we could find a means to wirelessly update the application, but that time may be better spent implementing as many features as possible. More user-friendly features are especially important, such as potential save file salvaging, universal undo function, etc. Ultimately, we fulfilled the bare minimum of our ambitions and gave ourselves little time to do anything else.